



Elastic Collisions in Two Dimensions

• Mark Scheme

Further Mechanics 1 Chapter 5: Elastic Collisions in Two Dimensions. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Oblique impact with a fixed surface

- 1 Perpendicular component becomes $0.5 \times 3 = 1.5 \text{ ms}^{-1}$, parallel unchanged M1; speed $\sqrt{4^2 + 1.5^2} = 4.27 \text{ ms}^{-1}$ A1.
- 2 Parallel: $10 \cos 60^\circ = 5$; perpendicular after: $0.4 \times 10 \sin 60^\circ = 3.46$ M1; speed $\sqrt{25 + 12} = 6.08 \text{ ms}^{-1}$ A1; angle $\tan^{-1} \left(\frac{3.464}{5} \right) = 34.7^\circ$ with the wall A1.
- 3 i-component unchanged: 5 B1; j-component reversed and multiplied by e : $+0.5 \times 4 = 2$ M1; $\mathbf{v} = (5\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$ A1.
- 4 Parallel $8 \cos 30^\circ = 4\sqrt{3}$, perpendicular $8 \sin 30^\circ = 4$ M1; perpendicular after = 2 A1; KE loss = $\frac{1}{2}(0.5)(4^2) - \frac{1}{2}(0.5)(2^2)$ (parallel part unchanged) M1; = 3 J A1.
- 5(a) Parallel component $10 \cos \alpha = 8$ (unchanged) B1; perpendicular $10 \sin \alpha = 6$ becomes $\frac{1}{3} \times 6 = 2$ M1; $\tan \beta = \frac{2}{8} = \frac{1}{4}$ A1.
- 5(b) $\sqrt{8^2 + 2^2}$ M1; = $\sqrt{68} = 8.25 \text{ ms}^{-1}$ A1.
- 5(c) Deflection = $\alpha + \beta = 36.87^\circ + 14.04^\circ$ M1; = 50.9° A1.
- 5(d) $\frac{\frac{1}{2}m(100) - \frac{1}{2}m(68)}{\frac{1}{2}m(100)}$ M1; = 0.32 A1.

Section B – Oblique impact of two smooth spheres

- 6 j-component unchanged: $\mathbf{v}_A = (-\mathbf{i} + 4\mathbf{j}) \text{ ms}^{-1}$ B1; speed $\sqrt{17} = 4.12 \text{ ms}^{-1}$ B1.
- 7 Along i: $2(4) + 1(-2) = 2a + b$ M1; $b - a = 0.5(4 + 2)$ M1; $a = 1, b = 4$: $\mathbf{v}_A = (\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$, $\mathbf{v}_B = (4\mathbf{i} + 3\mathbf{j}) \text{ ms}^{-1}$ A1.
- 8 Along the line of centres A has $6 \cos 30^\circ = 3\sqrt{3}$, perpendicular $6 \sin 30^\circ = 3$; equal masses with $e = 1$ exchange line-of-centres components M1; A: 3 ms^{-1} perpendicular to the line of centres A1; B: $3\sqrt{3} = 5.20 \text{ ms}^{-1}$ along the line of centres (so they move at right angles) A1.
- 9 Along line of centres: $m\left(\frac{1}{2}u\right) = ma + mb$ M1; $b - a = \frac{1}{2}\left(\frac{1}{2}u\right)$ M1; $a = \frac{1}{8}u, b = \frac{3}{8}u$, so speed of B is $\frac{3}{8}u$ A1; A: $\sqrt{\left(\frac{1}{8}u\right)^2 + \left(\frac{\sqrt{3}}{2}u\right)^2} = \sqrt{\frac{49}{64}}u = \frac{7}{8}u$ A1.
- 10(a) A: $5 \cos \alpha = 4$ along, $5 \sin \alpha = 3$ perpendicular B1; $2(4) + 3(-2) = 2a + 3b$ M1; $b - a = \frac{1}{2}(4 + 2) = 3$ M1; $a = -1.4 \text{ ms}^{-1}, b = 1.6 \text{ ms}^{-1}$ (both directions reversed) A1.
- 10(b) $\sqrt{1.4^2 + 3^2}$ M1; = 3.31 ms^{-1} A1; $\tan^{-1} \left(\frac{3}{1.4} \right) = 65.0^\circ$ with the line of centres A1.
- 10(c) Before $\frac{1}{2}(2)(25) + \frac{1}{2}(3)(4) = 31 \text{ J}$; after $\frac{1}{2}(2)(10.96) + \frac{1}{2}(3)(1.6^2) = 14.8 \text{ J}$ M1; loss = 16.2 J A1.

Roar-level challenge

- 11 Along line of centres: $a + b = u \cos 30^\circ, b - a = e u \cos 30^\circ$ M1; $a = \frac{1}{2}u \cos 30^\circ(1 - e)$, perpendicular component $u \sin 30^\circ$ unchanged A1; $\tan 60^\circ = \frac{u \sin 30^\circ}{\frac{1}{2}u \cos 30^\circ(1 - e)} \Rightarrow \sqrt{3} = \frac{2}{\sqrt{3}(1 - e)}$ M1; $e = \frac{1}{3}$ A1; after: $v_A^2 = \frac{1}{12}u^2 + \frac{1}{4}u^2 = \frac{1}{3}u^2$, $v_B^2 = \left(\frac{\sqrt{3}}{3}u\right)^2 = \frac{1}{3}u^2$, so KE after = $\frac{2}{3}$ of KE before: fraction lost $\frac{1}{3}$ B1.